Benchmarking Multi-View Tree-Level Oil Palm Bunch Counting Under a Fixed Detector

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Abstract—Tree-level yield forecasting often requires a count of objects in several readiness classes rather than a single total. Multi-view imaging helps recover objects hidden by occlusion, but it also creates duplicate observations of the same physical object across views. A deployed system must therefore solve detection and tree-level aggregation jointly. Whether failures arise from the detector or the aggregation step has not been isolated by prior work. This paper benchmarks that question on 953 oil palm trees and 3,992 side-view images under two detection conditions. The benchmark evaluates five regression counters under both conditions and ablates eight feature configurations in the fixed-detector setting. When the counting model receives annotated detections, it reaches 98.05% accuracy at the class level and 92.20% at the whole-tree level. When the same task uses outputs from a fixed detector (YOLO26m), those accuracies drop to 77.48% and 32.62%. The 20.57 percentage-point gap remains after testing all model and feature combinations. This indicates detector quality is the limiting factor, not the tree-level counting step.

Keywords—Oil palm, fresh fruit bunch, black bunch census, multi-view counting, object detection, YOLO, tree-level regression, agricultural computer vision.

I. Introduction

Harvest scheduling in commercial oil palm plantations depends on knowing, for every tree, how many fruit bunches belong to each operational maturity class. The Black Bunch Census (BBC) formalises this requirement into a four-class count vector, B1 through B4, where each class carries distinct timing and logistics implications [1]. Automating BBC through image-based counting would reduce the cost and subjectivity of manual field surveys. The central operational requirement is reproducing the per-class disaggregation rather than collapsing it into a single total.

Tree-level counting is geometrically harder than image-level detection [2]. Bunches surround the full circumference of an oil palm, so a single image misses bunches due to occlusion, viewing angle, or overlapping fronds. Multi-view acquisition reduces missed bunches by photographing each tree from four to eight side views, but the same physical bunch then appears in several images. Fig. 1 illustrates the effect: a single B3 bunch is visible in three adjacent side views of the same tree. Naive summation of per-image detections counts appearances rather than bunch identities and is biased upward on this dataset.

Multi-view deduplication has been studied across several crops through geometric fusion, sequential scan, and neural reconstruction. Mango counting systems associated detections across sequential passes using navigation data and canopy geometry [3], and apple pipelines built row-level maps from overlapping two-side scans [4]. Comparative evaluation of multi-view apple pipelines shows that counting errors typically originate in detection quality rather than the aggregation step [5]. Sparse and calibrated viewpoints have been addressed through correspondence-free triangulation [6] and few-view 3D reconstruction [7], and recent neural-field methods count from unstructured photographs in a shared 3D representation [8]. Most of these systems depend on dense sequential overlap, a moving sensor platform, or calibrated geometry, and none isolates detector quality from counter quality as separate error sources.

Regression-based aggregation maps detection statistics to a tree-level count vector, an approach well established in agricultural vision [2], [4], [5]. YOLO-style single-stage detectors are now the standard front-end for such pipelines [9], [10]. The regressor is typically evaluated only on the detector outputs available during the study, so detector quality and counter quality enter the accuracy estimate jointly. When end-to-end performance falls short, this coupling makes it impossible to determine whether effort should target the detector or the counter. Isolating the two contributions requires evaluating the same counting model under both controlled and realistic detection

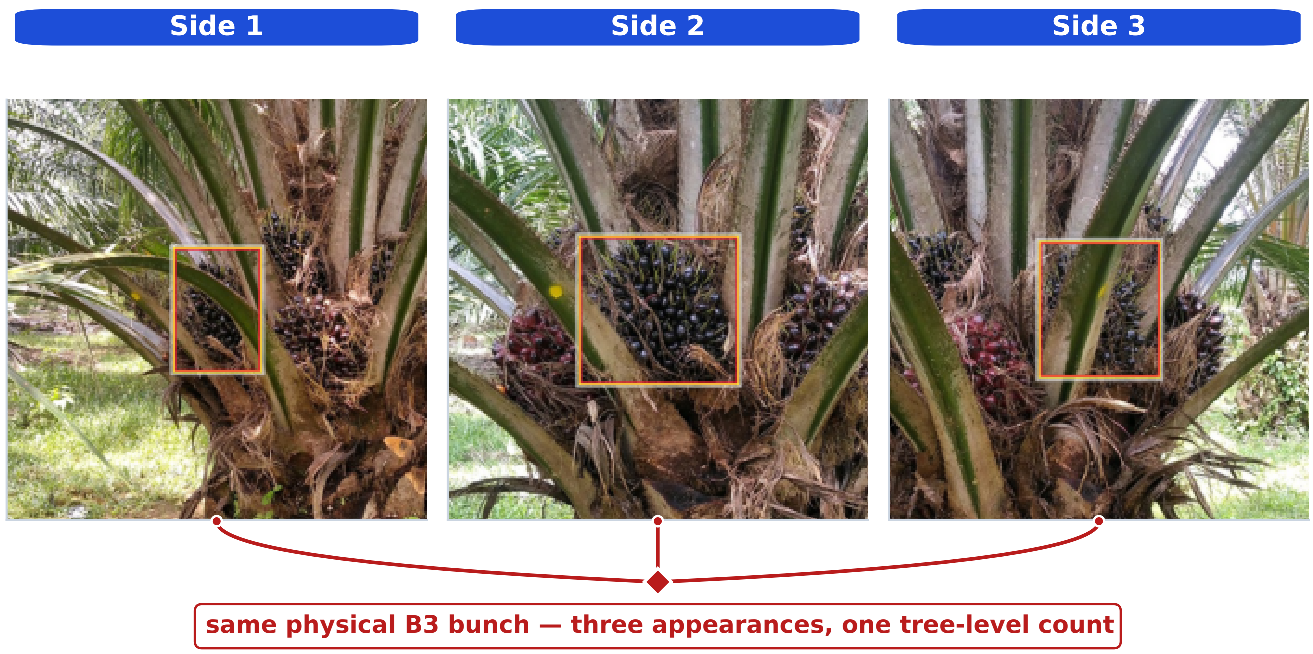

Fig. 1. A single physical B3 bunch is visible across three adjacent side views. If every per-image detection is counted, three appearances are added for one bunch. Tree-level counting must therefore aggregate evidence across views rather than sum it.

conditions.

Oil palm counting has received less systematic study than apple or mango. Detectors for fresh fruit bunches (FFB) and maturity classification have reached production-level accuracy on ripe bunches but show recurring weaknesses on partially occluded and maturity-ambiguous classes [15], with both two-stage and YOLO-style architectures explored [11]–[14]. The closest public oil palm resource is a video dataset annotated at the frame level for detection [16]. Tree-level counting labels are absent, and no counting evaluation has been reported on it. No existing benchmark evaluates tree-level counting strategies under controlled detection conditions or separates detector error from counter error for this task.

This paper closes that gap by benchmarking tree-level multi-view BBC counting under two detection conditions on the SawitMVC dataset (953 trees, 3,992 images) [17]. The first condition uses ground-truth detections to isolate counter performance from detector error, and the second uses a fixed YOLO26m detector to reflect a realistic deployed pipeline. Five regression models and eight feature configurations are evaluated within each condition. The paper makes three contributions: (1) a systematic benchmark of tree-level bunch counting under two detection conditions, establishing an accuracy ceiling and a deployment baseline on the same dataset; (2) a feature ablation study showing that enriched feature representations recover only a marginal fraction of the detector-driven gap; and (3) evidence that the primary performance bottleneck is detector output quality rather than the choice of counting model.

II. Methodology

A. Dataset and Task Definition

The benchmark uses the SawitMVC multi-view dataset: 953 oil palm trees, 3,992 side-view images, four maturity classes (B1, B2, B3, and B4), and 4 to 8 side views per tree. The fixed split contains 716 training trees, 96 validation trees, and 141 test trees. Dataset construction, annotation protocol, and per-class composition are documented in [17].

For tree i, the ground-truth target is the count vector

$$\mathbf{y}_i = [\,y_{i,B1},\ y_{i,B2},\ y_{i,B3},\ y_{i,B4}\,], \tag{1}$$

and the prediction is $\hat{\mathbf{y}}_i = [\,\hat{y}_{i,B1}, \ldots, \hat{y}_{i,B4}\,]$. The evaluation unit is the tree, not the individual image: per-image appearance counts are not valid BBC outputs.

Let $a_{i,c}$ denote the number of detected appearances of class c across all views of tree i. The naive multi-view estimator

$$\hat{y}^{\text{naive}}_{i,c} = a_{i,c} \tag{2}$$

is biased upward whenever a bunch is visible from more than one side. Under GT annotations, this estimator reaches only 50.00% Class ±1 Acc and 6.38% Tree ±1 Acc on the test split, which motivates learned tree-level aggregation rather than direct summation.

B. Detection Conditions

The benchmark isolates detector error from counter error by evaluating the same counters under two detection conditions, illustrated in Fig. 2.

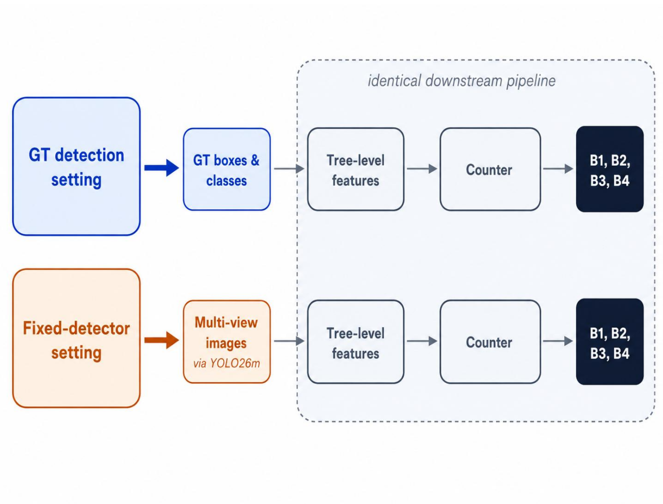

Fig. 2. Two detection conditions used in the benchmark. Top: the GT detection setting supplies ground-truth boxes and classes to the tree-level feature extractor, so detector output quality is removed as an error source. Bottom: the fixed-detector setting supplies cached YOLO26m outputs, so detector misses and class confusions enter the pipeline before counting.

In the GT detection setting, the feature extractor reads ground-truth bounding boxes and classes directly from the annotation files. Counters trained on these features upper-bound how well a tree-level counter can perform when detections are accurate. In the fixed-detector setting, the feature extractor reads cached YOLO26m [10] predictions from the released y26mv2 checkpoint. The detector was trained on the official 716-tree training split for 60 epochs (batch size 32, image size 640, patience 60, seed 42). Inference uses the repository default confidence threshold of 0.25 and the standard Ultralytics post-processing settings. A single detector checkpoint is used for all counter and feature configurations, so any observed differences come from the counter or feature representation rather than detector retraining. Preliminary experiments across multiple detector architectures and training configurations produced no meaningful performance differences on this dataset, indicating that the bottleneck lies in task difficulty rather than detector design. The y26mv2 checkpoint was therefore selected as a representative fixed detector. A deeper detector study, including alternative architectures and depth-augmented inputs, is planned as future work.

The per-class weakness on B4 and the moderate recall on B3 in Table I are revisited in Section III-D.

C. Feature Vectors

Given either set of detections, observations from all views of a tree are aggregated into a fixed-length feature vector. The 13-dimensional baseline F_0 contains, for each

TABLE I
YOLO26m validation detection performance, reported under the COCO mAP50 convention [18].

Class	mAP50	Recall
B1	0.746	0.801
B2	0.425	0.433
B3	0.550	0.656
B4	0.363	0.389
Overall	0.521	0.570

class c, the total count s_c, the per-side maximum m_c, and the per-side mean μ_c, plus the number of available views n_{views}:

$$F_0 = [\, s_{B1:B4},\ m_{B1:B4},\ \mu_{B1:B4},\ n_{\text{views}} \,]. \tag{3}$$

Three optional groups extend F_0. The confidence group (20-dim) contains five per-class statistics: confidence sum, mean, max, count above 0.5, and count above 0.6. Spatial position provides a separate signal: the mean normalized vertical centroid $\overline{cy}_c$ and mean bounding-box area $\overline{A}_c$ for each class form an 8-dimensional group that reflects the vertical separation of maturity classes on the tree. A 20-dimensional side-distribution group records per-side std, per-side min, coefficient of variation, number of sides with detections, and a consistency score for each class.

A cross-class composition group (6-dim), consisting of the total detection count, four class fractions, and a B3-vs-(B2+B3) mixture ratio, completes the 67-dimensional combined set F_{all}:

$$F_{\text{all}} = F_0 \cup \text{conf} \cup \text{spatial} \cup \text{distrib} \cup \text{composition}.$$

Eight feature sets are evaluated: F_0, F_0+conf, F_0+spatial, F_0+distrib, F_0+conf+spatial, F_0+conf+distrib, F_0+distrib+spatial, and F_{all}.

D. Counter Models and Evaluation Metrics

Three heuristic baselines are included as reference points. The naive sum sums all detected appearances per class (defined in Section II-A). The global divisor scales the naive sum by a fixed constant k fit to the training split, correcting for the average number of times a single bunch is observed across views. The visibility-adaptive divisor replaces the global constant with a per-tree factor that varies with the number of side views captured for that tree.

Five regression models are evaluated in both detection conditions: Linear Regression, Ridge [19], ElasticNet [20], SVM [21], and Random Forest [22]. Each counter maps a tree feature vector $\mathbf{x}_i$ to the count vector $\hat{\mathbf{y}}_i = f_\theta(\mathbf{x}_i)$.

All counter models are trained on the 716-tree training split and evaluated on the fixed 141-tree test split. Class-level ± 1 correctness for tree i and class c is

$$I^{\pm 1}_{i,c} = \begin{cases} 1, & |\hat{y}_{i,c} - y_{i,c}| \le 1, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Class ±1 Acc averages this indicator over all test trees and classes:

$$\text{Class } \pm 1 \text{ Acc} = \frac{1}{4N} \sum_{i=1}^{N} \sum_{c=1}^{4} I_{i,c}^{\pm 1}. \tag{5}$$

Tree ±1 Acc is stricter: a tree counts as correct only if all four class predictions are within ±1 simultaneously,

$$\text{Tree } \pm 1 \text{ Acc} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}\left(\sum_{c=1}^{4} I_{i,c}^{\pm 1} = 4 \right). \tag{6}$$

Macro MAE is the mean absolute error averaged over trees and classes. Signed class bias is the per-class mean $\hat{y}_{i,c} - y_{i,c}$; positive values indicate over-counting, negative values under-counting. Bias is reported alongside accuracy because operational aggregate yield estimates are sensitive to directional error rather than absolute error alone.[1]

III. Results and Discussion

The results follow the detector-counter question directly. Section III-A estimates the counter ceiling under GT detections, Section III-B measures the deployed fixed-detector setting, Section III-C tests whether richer features recover the loss, and Section III-D compares the two conditions per class.

A. GT Detection Setting

Table II reports counting results when the counter receives GT detections. Naive appearance summation yields only 50.00% Class ±1 Acc and 6.38% Tree ±1 Acc. This confirms that duplicate visibility cannot be ignored and that direct summation is not a viable estimator. The global divisor and visibility-adaptive divisor recover most of the loss, and ElasticNet improves further to 98.05% Class ±1 Acc. Under accurate detections, the remaining counting problem is therefore simple enough for compact regression models.

TABLE II
Counting under the GT detection setting on the 141-tree test split.

Method	Type	Class ±1	Tree ±1	MAE
Naive sum	Heur.	50.00%	6.38%	2.142
Global divisor	Heur.	95.39%	85.11%	0.376
Visibility-adapt. divisor	Heur.	95.92%	87.23%	0.340
ElasticNet	ML	98.05%	92.20%	0.277
SVM	ML	97.87%	91.49%	0.266
Ridge	ML	97.70%	90.78%	0.275
Linear reg.	ML	97.52%	90.07%	0.277
Random forest	ML	95.92%	84.40%	0.365

The four best machine-learning counters (ElasticNet, SVM, Ridge, LR) sit within 0.5 percentage points of each other on Class ±1 Acc and within 2.2 points on Tree ±1 Acc. Random Forest is the only counter that underperforms in this setting. This narrow spread indicates that, when detector evidence is accurate, the choice among standard counter families is not the limiting factor.

[1]Source code: https://github.com/ULM-SawitMVC/Baseline-SawitMVC.

B. Fixed-Detector Setting

Moving from GT to cached YOLO26m detections, the best F_0 result (ElasticNet, 76.42% Class ±1 Acc) trails the same counter under GT by more than 20 percentage points (Table III). All five counters remain within a 3.2 percentage-point window on Class ±1 Acc. The large drop across the two detection conditions, coupled with the small spread among counters, points to the detector output rather than the counter family as the main source of loss.

TABLE III
Fixed-detector counting with F0 features on the 141-tree test split.

Counter	Class ±1 Acc	Tree ±1 Acc	Macro MAE
ElasticNet	76.42%	29.79%	1.043
Ridge	76.06%	28.37%	1.053
Linear Regression	75.71%	30.50%	1.048
SVM	74.82%	29.08%	1.043
Random Forest	73.23%	26.95%	1.110

The per-class breakdown in Table IV shows that B1 stays near its GT ceiling while B3 falls to 55–56% Class ±1 Acc across all five counters. B2 and B4 sit at intermediate losses. This profile matches the detection recall in Table I: the weakest detector classes lose the most counting accuracy.

TABLE IV
Per-class Class ±1 Acc of fixed-detector counters with F0 features on the 141-tree test split.

Counter	B1	B2	B3	B4
ElasticNet	96.45%	80.14%	56.03%	73.05%
Ridge	95.74%	80.14%	56.03%	72.34%
Linear Regression	96.45%	79.43%	55.32%	71.63%
SVM	95.74%	76.60%	55.32%	71.63%
Random Forest	95.74%	73.76%	56.03%	67.38%

C. Feature Ablation

TABLE V
Ridge feature ablation in the fixed-detector setting.

Features	Dim	Class ±1	Tree ±1	MAE
F_0	13	76.06%	28.37%	1.053
F_0 + confidence	33	75.89%	29.79%	1.059
F_0 + spatial	21	76.60%	30.50%	1.046
F_0 + side-distribution	33	74.82%	28.37%	1.060
F_0 + confidence + spatial	41	76.24%	29.08%	1.059
F_0 + confidence + side-distribution	53	76.42%	29.79%	1.060
F_0 + side-distribution + spatial	41	75.71%	30.50%	1.048
F_{all}	67	77.48%	32.62%	1.035

Whether enriched features can recover the detector-driven gap is tested by fixing Ridge and varying the feature configuration (Table V). F_0 starts at 76.06% Class ±1 Acc. Spatial features alone raise this to 76.60%; the vertical centroid and bounding-box area encode residual maturity position that the raw count statistics in F_0 do not capture directly. Confidence features do not improve the baseline individually, and adding side-distribution alone reduces Class ±1 Acc to 74.82%, below the F_0 baseline, suggesting this group introduces noise under the current detector.

The full F_{all} set gives the best ablation result at 77.48% Class ±1 Acc and 32.62% Tree ±1 Acc. The gain over F_0 is 1.42 percentage points at class level and 4.25 points at tree level. This 1.42-pp gain is the ceiling of feature-side improvement. The best feature set narrows only a small fraction of the gap between GT and fixed-detector conditions.

D. GT vs. Fixed-Detector Gap

The central comparison is the per-class gap between ideal detections and the fixed detector, shown with the signed bias of the best configuration in Fig. 3.

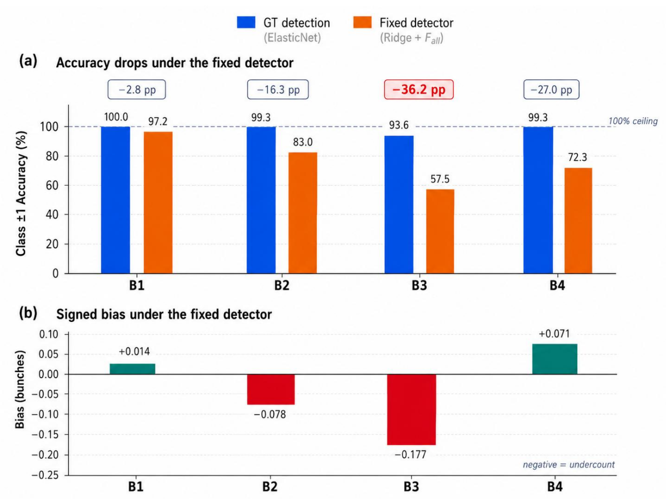

Fig. 3. Top: Per-class Class ±1 Acc for the best counter in each detection condition (ElasticNet under GT; Ridge + F_{all} under the fixed detector). Counting is near ceiling under GT detection but drops sharply under the fixed detector. Bottom: Per-class signed bias of Ridge + F_{all} in the fixed-detector setting. Negative values indicate systematic under-counting; the under-count is concentrated on B2 and especially B3.

The absolute gap is large, at 20.57 percentage points in Class ±1 Acc and 59.58 percentage points in Tree ±1 Acc, as summarized in Table VI. The loss is not uniform across maturity classes. B1 loses only ≈2.8 pp between the two conditions, while B2 loses ≈16.3 pp, B3 loses 36.2 pp, and B4 loses ≈27.0 pp. This profile aligns with Table I, where B3 and B4 have weaker detector recall and fall furthest below the GT ceiling.

The bias plot adds direction to the accuracy gap. The fixed-detector Ridge + F_{all} pipeline under-counts B2 by 0.078 and B3 by 0.177 bunches per tree on average. Because plantation-level forecasts aggregate per-tree predictions across blocks, systematic under-counting in B2 and B3 would carry into the next-cycle harvest estimate rather than cancel out as random noise.

Across all counter and feature combinations, the per-class gap traces the detector recall profile in Table I, not

TABLE VI
Consolidated test-set summary. Biases are ordered by class.

Config	Class ±1	Tree ±1	B1	B2	B3	B4
GT, ElasticNet	98.05%	92.20%	−.050	+.043	−.064	−.028
Fixed, Ridge + F_{all}	77.48%	32.62%	+.014	−.078	−.177	+.071

the spread among counter families. Multi-view duplicate visibility is real and large, but learned tree-level counters handle it when detector evidence is accurate (Section III-A). Once a fixed detector replaces ground-truth boxes, counter performance drops sharply (Section III-B), and richer features or different counter families close only a small fraction of the loss (Section III-C). This aligns with the finding in [5] that counting errors typically trace to detection quality. The present benchmark puts a concrete upper bound on what counter-side improvements can contribute.

These findings have a direct consequence for system design. The gap between the GT ceiling and the fixed-detector result sets an upper bound on what counter-side changes alone can recover. No counter or feature combination tested here narrows that gap by more than 1.42 pp at the class level. Improving detector recall on B3 and B4, the two weakest classes, is therefore the highest-leverage intervention available in this pipeline.

Several limitations bound the scope of these findings. The benchmark uses a single dataset of 953 trees from two plantations in South Kalimantan, so absolute numbers may shift under other regions, varieties, or acquisition protocols. Only one detector checkpoint (YOLO26m) is tested, and a different detector family could change the per-class error profile. The tree-level counters treat each tree independently and ignore plantation-scale spatial or temporal regularities that operational forecasts sometimes exploit. The four-class B1 through B4 taxonomy also admits visual ambiguity at the B2/B3 and B3/B4 boundaries. Future work should prioritize B3/B4 recall, confidence calibration for counting, and explicit cross-view association or geometry-aware aggregation [6], [7], [23] rather than expanding the tree-level feature set further.

IV. Conclusion

This paper benchmarks multi-view tree-level oil palm FFB counting by evaluating the same counters under GT detections and cached YOLO26m outputs. With GT detections, standard regression counters achieve near-ceiling accuracy on the B1 through B4 count. With the fixed detector, neither additional feature groups nor alternative counter families narrow the gap appreciably. On SawitMVC under this detector checkpoint, the limiting factor is detector output quality, especially for B3 and B4, rather than the choice among the tested tree-level counters. Closing the gap will require improving the detector, not the counter.

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